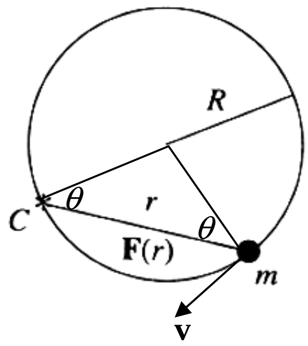
3. A particle of mass m moves in a circular orbit of radius R under the influence of a central force $F(r)$. The center of the central force lies at a point on the circle, as shown in the accompanying figure. What is the force law of the central force? (Hint: the equation of the orbit can be written as $r = 2R\cos\theta$.)



Solution: The equation of the circular orbit can be written as

$$r = 2R\cos\theta, \quad \cos\theta = \frac{r}{2R}.$$

where R is the radius of the circle and r is the distance of the mass to the center of the force.
The equation of motion is

$$F(r)\cos\theta = m\frac{v^2}{R}.$$

The angular momentum with respect to the center of the force is conserved:

$$L = m|\vec{r}\times\vec{v}| = mrv\cos\theta = \text{Const.}$$

Then $v = \dfrac{L}{mr\cos\theta}$. Thus

$$F(r) = m\frac{v^2}{R\cos\theta} = \frac{L^2}{mRr^2\cos^3\theta} = \frac{2L^2}{mr^3\cos^2\theta} = \frac{2L^2}{mr^3\left(\dfrac{r}{2R}\right)^2} = \frac{8R^2L^2}{mr^5}.$$

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The force law can also be determined by using the Binet equation

$$F(u) = -mh^2u^2\left(\frac{d^2u}{d\theta^2} + u\right)$$

where $u = \dfrac{1}{r}$ (the reciprocal of r) and $h = \dfrac{L}{m}$.

The derivation of the Binet equation is as follows.

The Newton's second law for the central force is

$$F(r) = m\left(\ddot{r} - r\dot{\theta}^2\right)$$

and the conservation of angular momentum can be written as

$$r^2\dot{\theta} = \frac{L}{m} = h = \text{constant}.$$

Then

$$\frac{du}{d\theta} = \frac{d}{dt}\left(\frac{1}{r}\right)\frac{dt}{d\theta} = -\frac{\dot{r}}{r^2\dot{\theta}} = -\frac{\dot{r}}{h},$$

$$\frac{d^2u}{d\theta^2} = -\frac{1}{h}\frac{d\dot{r}}{dt}\frac{dt}{d\theta} = -\frac{\ddot{r}}{h\dot{\theta}} = -\frac{\ddot{r}}{h^2u^2}.$$

Finally

$$F(r) = m\left(\ddot{r} - r\dot{\theta}^2\right) = -m\left(h^2u^2\frac{d^2u}{d\theta^2} + h^2u^3\right) = -mh^2u^2\left(\frac{d^2u}{d\theta^2} + u\right).$$